The Frohmanian Tether on SDiff($\mathbb{T}^3$): A Metriplectic Correction, Its Canonicity, and an Obstruction to Damping

Conditional statements verified in Lean 4

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Companion repository: https://github.com/BenFrohman/frohmanian-tether-sdiff (version 0.1.1)

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Abstract

We describe a Lean 4/Mathlib formalization of the *Frohmanian tether*, a symmetric quadratic correction $T(F,G) = -\kappa \|\omega\|_{L^2}^2 \int \langle \Pi_u \, \delta F/\delta\omega, \Pi_u \, \delta G/\delta\omega \rangle$ added to the Arnold Lie–Poisson bracket of ideal fluids, where Π_u is the L^2-orthogonal projection away from the velocity u and $\kappa = 3/2$. The library proves: the negative-feedback condition (C3) unconditionally; the kinetic-energy degeneracy (C2) and coadjoint invariance (C1) under explicitly named hypotheses; a uniqueness ("canonicity") statement inside a restricted kernel class; and an *obstruction to damping*: for every symmetric kernel satisfying (C2), $T(M,H) = T(H,M) = 0$ for every functional M, so the tether contributes no damping to any functional, in particular not to $\|\omega\|_{L^\infty}$. On the analytic side the library proves a scalar comparison lemma for $y' \le Cy^2 - \kappa'' y^3$ and several conditional implications of the form "(cubic rate or uniform vorticity bound) + (Kato local existence) + (Beale–Kato–Majda continuation) + (uniqueness) $\Rightarrow$ global smooth solution", in which every analytic input is an explicit hypothesis. The rate and closing-bound hypotheses are open, are not supplied by the tether (its damping coefficient is 0), and are of the same strength as the global regularity problem. **This work does not prove global regularity of the three-dimensional Navier–Stokes equations and makes no claim toward the Clay Millennium Problem.** The build (Lean `v4.30.0-rc1`, Mathlib `v4.30.0-rc1`, 2712 jobs) contains 519 theorem constants (483 stated in the sources and 36 auto-generated equation, congruence and injectivity lemmas) and 761 declarations in 22 modules, with no `sorry`, no `native_decide`, no axiom declarations, and every declaration depending only on `propext`, `Classical.choice` and `Quot.sound`; this is enforced in continuous integration.

Contents

1 Introduction and scope **2**

2 The formal setting **3**

3 The tether kernel and conditions (C1)–(C3) **3**

3.1 The identification $\delta H/\delta\omega = u$ 4

4 The obstruction to damping **4**

5 Canonicity **5**

6 Bracket structure and Casimirs **5**

7 The analytic layer **6**

8 The combined capstone **7**

9 The periodic Clay statement and a conditional reduction **7**

10 What the Lean development does and does not prove **8**

A Exact Lean statements **8**
A.1 Setting 8
A.2 Tether, (C1)–(C3), obstruction, canonicity 9
A.3 Analytic layer 14
A.4 Periodic Clay statement 18

B Verification certificate **19**

1 Introduction and scope

Arnold's geometric picture of ideal hydrodynamics [1, 2] describes the Euler equations as Hamiltonian motion on coadjoint orbits of the group SDiff of volume-preserving diffeomorphisms, with Lie–Poisson bracket $\{F, G\}_{\mathrm{cl}}(\omega) = \int \omega \cdot (\delta F/\delta\omega \times \delta G/\delta\omega)$ and kinetic-energy Hamiltonian $H(\omega) = \frac{1}{2}\int |\mathrm{BS}\,\omega|^2$. The *Frohmanian tether* programme adds to this bracket a symmetric, negative-semidefinite quadratic term T and asks whether T can supply a damping mechanism for the vorticity sup-norm $M(t) = \|\omega(t)\|_{L^\infty}$, which by the Beale–Kato–Majda criterion [3] controls continuation of smooth Navier–Stokes solutions.

This paper reports exactly what the accompanying Lean 4 library proves. The Lean statements are the source of truth: every result below is stated to match a checked Lean declaration, with every hypothesis made explicit, and the exact Lean text is reproduced in Appendix A. The main findings are:

(i) The tether T satisfies (C3) unconditionally, (C2) under the identification $\delta H/\delta\omega = u$ together with $\nabla \cdot u = 0$ and $u \in L^2$, and (C1) under two named invariance hypotheses (Section 3).

(ii) *Obstruction to damping.* For every symmetric kernel B satisfying (C2), $B(H, M) = 0$ for every functional M; for T this gives $T(M, H) = T(H, M) = 0$ (Section 4). The tether therefore cannot produce the cubic damping rate on which the programme relied.

(iii) *Canonicity* holds only inside a restricted class and follows from saturation and polarization hypotheses; conditions (C1)–(C3) appear in the statement but are not used (Section 5).

(iv) The analytic layer consists of proved real-variable lemmas and *conditional* implications whose open hypotheses (a cubic rate or a uniform vorticity bound for all data, Kato local existence, BKM continuation, uniqueness) are of Clay strength (Sections 7–9).

Scope statement. No result in this paper or in the library asserts global existence, smoothness or uniqueness of solutions of the three-dimensional Navier–Stokes equations. Every theorem whose conclusion mentions global solutions is an implication from hypotheses that are open. Section 10 lists precisely what the Lean development does and does not prove.

Version note. Version 0.1.0 of this manuscript named a different repository (`NS_Millennium_Proof`), quoted theorem and job counts of that repository, and cited a Zenodo DOI belonging to it. Version 0.1.1 refers only to the standalone repository above and uses the counts produced by its own build (Appendix B). No DOI is assigned to this version in the manuscript.

2 The formal setting

We record the definitions as formalized, including the points at which they differ from the classical objects.

- **Space.** `T3` is `EuclideanSpace ℝ (Fin 3)`, i.e. $\mathbb{R}^3$ with Lebesgue measure, *not* the quotient torus $\mathbb{R}^3/\mathbb{Z}^3$. Periodicity, where needed (Section 9), is the predicate `IsZ3Periodic` on functions on $\mathbb{R}^3$. The notation SDiff($\mathbb{T}^3$) in the title refers to the motivating geometry; no diffeomorphism group of $\mathbb{T}^3$ is constructed.
- **Phase space.** `CoadjointOrbit` is the type of pointwise divergence-free vector fields $\omega : \mathbb{R}^3 \to \mathbb{R}^3$ (a subtype; no orbit structure is built in).
- **Functional derivative.** `FunctionalDerivative F ω` is a *choice* of a Gâteaux representative δ of F at ω along C^1 divergence-free variations, and is 0 if none exists. A Gâteaux representative is in general determined only up to the pairing against admissible variations, so the chosen field is not canonical.
- **Velocity and energy.** `velocity_from_vorticity ω` is the Biot–Savart integral `BiotSavart ω.val`, and `KineticEnergyHamiltonian ω` $= \frac{1}{2}\int \|\mathrm{BS}\,\omega\|^2$.
- **Navier–Stokes.** `navier_stokes_eq u p ν` (and `NS_PDE`) asserts the momentum equation with Mathlib's total derivative `deriv` in time, and $\nabla \cdot u = 0$, pointwise for $t \geq 0$. `IsPhysicalNSSolution u` adds pointwise time-differentiability for $t \geq 0$, spatial continuity, finite kinetic energy, the energy inequality $\int \|u(t)\|^2 \leq \int \|u(0)\|^2$, and $u(t) = u(0)$ for $t \leq 0$.

The exact definitions are quoted in Appendix A.1.

3 The tether kernel and conditions (C1)–(C3)

Definition 3.1 (`Pi_u`, `TetherKernel`, `TetheredBracket`)**.** For velocity fields u, v put $\Pi_u v = v - \left(\int \langle v, u\rangle / \int \|u\|^2\right) u$ (in Lean, division by 0 is 0, so $\Pi_u v = v$ when $\int \|u\|^2 = 0$). With $\kappa = 3/2$ (`kappa_eq_three_halves`), $\mathcal{E}(\omega) = \int \|\omega\|^2$ (`Enstrophy`) and $u = \mathrm{BS}\,\omega$,

$$T(\omega; F, G) = -\kappa\,\mathcal{E}(\omega) \int \left\langle \Pi_u\,\delta F/\delta\omega,\ \Pi_u\,\delta G/\delta\omega \right\rangle dx, \qquad \{F, G\}_{\mathfrak{T}} = \{F, G\}_{\mathrm{cl}} + T(F, G).$$

The three conditions are predicates on a kernel $B(\omega; F, G)$: (C1) `InvariantUnderCoadjointAction`: $B(g\cdot\omega; F, G) = B(\omega; F, G)$ for all g in a group `SDiff` acting on `CoadjointOrbit`; (C2) `DegenerateWRTKineticEnergy`: $B(\omega; F, H) = 0$ for all F, ω; (C3) `ProducesControllableNegativeFeedback`: $B(\omega; F, F) \leq -\kappa\,\mathcal{E}(\omega) \int \|\Pi_u\,\delta F/\delta\omega\|^2$.

Theorem 3.2 (Symmetry and (C3); `tetherKernel_comm`, `tetherKernel_C3`)**.** *For all ω, F, G: $T(\omega; F, G) = T(\omega; G, F)$, and T satisfies (C3). No hypotheses.*

Theorem 3.3 ((C2), pointwise form; `tetherKernel_degenerates_on_kinetic_energy`)**.** *Let ω be such that `FunctionalDerivative KineticEnergyHamiltonian ω` $= \mathrm{BS}\,\omega$, $\nabla \cdot (\mathrm{BS}\,\omega) = 0$ pointwise, and $\|\mathrm{BS}\,\omega\|^2$ is integrable. Then $T(\omega; F, H) = 0$ for every F.*

Theorem 3.4 ((C2), global form; `Combined.tetherKernel_C2_of_KEDeriv`)**.** *Assume* `KineticEnergyDerivIsVelocity` *(Hypothesis 3.7), and that for every* ω, $\nabla\cdot(\mathrm{BS}\,\omega) = 0$ *pointwise and* $\|\mathrm{BS}\,\omega\|^2$ *is integrable. Then* T *satisfies (C2).*

Theorem 3.5 ((C1); `tether_coadjoint_invariance`, `Combined.tetherKernel_C1_of_casimir`)**.** *Let a group* `SDiff` *act on* `CoadjointOrbit`*. Assume* `SDiffActionHyp SDiff` *and* `TetherIntegralPreserved SDiff`*. Then* $T(g\cdot\omega; F, G) = T(\omega; F, G)$ *for all* g, F, G, ω*. The same conclusion holds with* `TetherIntegralPreserved` *replaced by a* `VolumePreservingCoadjointAction` *structure (*`Combined.tetherKernel_C1_of_volumePreserving`*).*

Remark 3.6 (On the hypotheses of Theorem 3.5). `SDiffActionHyp` is a structure with two fields: `casimir_invariant` (every Casimir $\int f(|\omega|)$ is invariant under the action) and `transitive_on_casimir_levels` (fields with equal Casimirs lie in one orbit). The proof uses only the first field (to obtain invariance of the enstrophy weight), but the theorem as stated requires the whole structure. The source documentation records that the transitivity field is not expected to hold for the genuine SDiff action, whose orbits are finer than Casimir level sets. `TetherIntegralPreserved` (invariance of the projected integral) is open. For the concrete group `ConcreteSDiff.SDiffConcrete` of measure-preserving, measurable, divergence-preserving permutations of $\mathbb{R}^3$ acting by $g\cdot\omega = \omega\circ g^{-1}$ (scalar composition, not the geometric push-forward $(Dg\,\omega)\circ g^{-1}$), the field `casimir_invariant` is proved (`ConcreteSDiff.casimir_invariant`); transitivity remains a hypothesis (`ConcreteSDiff.sdiffActionHyp_of_transitive`).

3.1 The identification $\delta H/\delta\omega = u$

Hypothesis 3.7 (`PlanSteps.KineticEnergyDerivIsVelocity`)**.** For every ω, `FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω`.

This is a named proposition, not a theorem and not an axiom. Its role in the library is exactly the following: it is a hypothesis of Theorem 3.4 and of the combined capstone (Theorem 8.1); its pointwise instance is a hypothesis of Theorem 3.3, of `tethered_reproduces_classical_euler`, and of the first clause of `frohmanian_tether_theorem` (Theorem 7.5). No theorem in the library proves it, and no theorem refutes it.

We believe it is *false as stated*. The Gâteaux derivative of $\frac{1}{2}\int|\mathrm{BS}\,\omega|^2$ in the direction ε is $\int\langle\mathrm{BS}\,\omega, \mathrm{BS}\,\varepsilon\rangle = \int\langle\mathrm{BS}^*\mathrm{BS}\,\omega, \varepsilon\rangle$, so a natural representative is $\mathrm{BS}^*\mathrm{BS}\,\omega$, which in the periodic setting is (up to gauge) the stream function ψ with $u = \nabla\times\psi$, not the velocity u. Moreover `FunctionalDerivative` selects one representative by choice, so even an identity that held up to the natural equivalence of representatives would not by itself give the pointwise equality of fields demanded above. If the hypothesis is false, then Theorem 3.4 and Theorem 8.1 are true but vacuous, and Theorem 3.3 gives information only at those ω (if any) at which its first hypothesis holds. Repairing this would require changing either the bracket's convention for $\delta H/\delta\omega$ or the projection Π_u (e.g. projecting away from $\mathrm{BS}^*\mathrm{BS}\,\omega$); we do not do so here.

4 The obstruction to damping

Theorem 4.1 (`Combined.left_H_zero_of_C2_symm`)**.** *Let* B *be any kernel that is symmetric,* $B(\omega; F, G) = B(\omega; G, F)$*, and satisfies (C2). Then* $B(\omega; H, M) = 0$ *for every* ω *and every functional* M*.*

Corollary 4.2 (`Combined.tetherKernel_left_H_zero`)**.** *If* T *satisfies (C2), then* $T(\omega; H, M) = T(\omega; M, H) = 0$ *for every* ω *and* M*.*

Theorem 4.1 is unconditional. For the tether, Corollary 4.2 requires (C2) as a hypothesis, which in turn is available only through Theorem 3.4, i.e. under Hypothesis 3.7. The interpretation is the following. Along the tethered flow $\dot{M} = \{M, H\}_{\mathfrak{T}} = \{M, H\}_{\mathrm{cl}} + T(M, H)$, the tether term vanishes; hence the symmetric design that makes the tether preserve the Euler vector field (C2) also forces its contribution to the evolution of every functional, including smoothed versions of $\|\omega\|_{L^\infty}$, to be zero. The negative diagonal values $T(F, F) \le 0$ of (C3) never enter dynamics generated by H. Consequently the cubic rate $M' \le CM^2 - \kappa'' M^3$ used below is not a consequence of the tether: its damping coefficient is 0.

Two related formal facts support this picture. `PlanSteps.tether_bound_not_scale_invariant`: for $\kappa > 0$, $M \neq 0$ and any real S, the family of inequalities $\lambda^6 S \le C(\lambda^2 M)^3 - \kappa(\lambda^2 M)^4$ $(\lambda > 0)$ fails for some λ, so a pointwise bound of this shape cannot be invariant under Navier–Stokes scaling. `tethered_stretching_of_weight`: the stretching pairing at a maximum plus the diagonal value $T(F, F)$ is at most $(CM - \kappa M^2)M^2$, but only under the weight hypothesis $M^4 \le \mathcal{E}(\omega) \int \|\Pi_u \, \delta F/\delta\omega\|^2$, for which no source is known.

5 Canonicity

Theorem 5.1 (`uniqueness_of_minimal_tether`; delegates to `uniqueness_of_minimal_symmetric_tether`)**.** *Let a group act on* `CoadjointOrbit` *and assume* `SDiffActionHyp`*. Let B be symmetric, satisfy (C1), (C2), (C3),* `SaturatesTetherQuadratic B` *(i.e. $B(\omega; F, F) = T(\omega; F, F)$ for all F, ω),* `Polarizes B` *and* `Polarizes TetherKernel` *(the polarization identity $2B(F, G) = B(F+G, F+G) - B(F, F) - B(G, G)$). Then $B = T$.*

Remark 5.2. The proof uses only saturation and the two polarization hypotheses: a kernel that agrees with T on the diagonal and satisfies polarization equals T. The hypotheses `SDiffActionHyp` and (C1)–(C3) are not used. Theorem 5.1 is therefore a uniqueness statement *given* agreement on the diagonal, not a derivation of T from (C1)–(C3).

Theorem 5.3 (`uniqueness_of_minimal_tether_core`)**.** *Let B be symmetric and assume: $B(\omega; F, G)$ depends on F only through $\Pi_u \, \delta F/\delta\omega$ (`h_proj`); if B factors through the projection then it has a density α (`HasTetherKernelDensity`) (`h_dens`); every such density is of the form $\alpha(\omega, x) = -c\,\mathcal{E}(\omega)$ (`h_weight`); B saturates T on the diagonal (`h_sat`); and there are ω, F with $\mathcal{E}(\omega) \int \langle \Pi_u \delta F/\delta\omega, \Pi_u \delta F/\delta\omega \rangle \neq 0$ (`h_nondeg`). Then $B = T$.*

6 Bracket structure and Casimirs

- `classicalBracket_antisymm`: $\{F, G\}_{\mathrm{cl}} = -\{G, F\}_{\mathrm{cl}}$; `classicalBracket_H_H`: $\{H, H\}_{\mathrm{cl}} = 0$. The Jacobi identity of $\{\cdot, \cdot\}_{\mathrm{cl}}$ is classical [1] and is *not* formalized.
- Since T is symmetric and in general nonzero, $\{\cdot, \cdot\}_{\mathfrak{T}}$ is *not* antisymmetric; it is of metriplectic type in the sense of Morrison [7] (antisymmetric part plus symmetric negative-semidefinite part). No Jacobi identity is claimed for it.
- `tethered_jacobi_identity` is a conditional lemma: if the Jacobiator of $\{\cdot, \cdot\}_{\mathfrak{T}}$ equals a cyclic Lie-pairing integral (`hident`) and that integral vanishes (`hvanish`), then the Jacobiator vanishes. Both hypotheses are open; the lemma does not establish a Jacobi identity.
- `PlanSteps.casimir_conserved`: under `SDiffActionHyp` and `IsCoadjointMotion` (the trajectory is $\omega(t) = g(t) \cdot \omega(0)$, an open hypothesis), every Casimir is constant along the trajectory. That the Euler flow is such a motion is not proved.
- `ScalingIdentities.vorticity_rescale`: for $u_\lambda(x) = \lambda u(\lambda x)$, $\nabla \times u_\lambda(x) = \lambda^2 (\nabla \times u)(\lambda x)$ under differentiability at λx.

7 The analytic layer

Lemma 7.1 (Comparison; `AnalyticPipeline.comparison_ode_stability`)**.** *Let $C, \kappa'' > 0$ and let $y : \mathbb{R} \to \mathbb{R}$ be differentiable at every $s \geq 0$ with $y'(s) \leq Cy(s)^2 - \kappa'' y(s)^3$ for $s \geq 0$. Then $y(t) \leq \max(y(0), C/\kappa'')$ for every $t \geq 0$.*

Proposition 7.2 (`riccati_ceiling_of_vorticity_di`)**.** *If $t \mapsto M(t) = \sup_x \|(\nabla \times u(t))(x)\|$ is continuous, differentiable at every $s \geq 0$, and satisfies $M'(s) \leq CM(s)^2 - \kappa'' M(s)^3$ for $s \geq 0$ with $C, \kappa'' > 0$, then M is bounded on $[0, \infty)$.*

The following hypotheses are named propositions (exact text in Appendix A.3); all are open for general data:

- `FrohmanianTetherRate u₀ ν` (R): for $\nu > 0$ and every physical solution from u_0, M is continuous, differentiable on $[0, \infty)$, pointwise bounded in space, and satisfies the cubic inequality for some $C, \kappa'' > 0$. (Differentiability of a sup-norm in time is itself a strong requirement.)
- `FrohmanianTetherBound u₀ ν`: every physical solution from u_0 has $\sup_{t \geq 0} M(t) < \infty$ (with continuity and spatial boundedness). `ClayStatements.FrohmanianTetherClosingBound ν` asserts this for *all* smooth divergence-free finite-energy data.
- `KatoLocalWitness u₀ ν`: data of a local smooth physical solution on $[0, T]$, $T > 0$ (local existence [6] is not formalized; a witness is an input).
- `PhysicalBKMContinuationCriterionFor u₀ ν`: the Beale–Kato–Majda criterion [3] for physical solutions from u_0, stated via the maximal time `Tstar`; not formalized, an input.
- `KatoUniquenessOnExistenceTimes u₀ ν w`: every smooth physical solution from u_0 agrees with the witness w on each existence interval; an input.

Theorem 7.3 (`ClosureBlueprint.conditional_regularity_of_open_hyps`)**.** *Let $\nu > 0$ and u_0 smooth, divergence-free, with $\|u_0\|^2$ integrable. Assume `KatoLocalWitness u₀ ν`, `FrohmanianTetherBound u₀ ν`, `PhysicalBKMContinuationCriterionFor u₀ ν` and `KatoUniquenessOnExistenceTimes u₀ ν w`. Then there exist u, p with `NS_PDE u p ν`, $u(0) = u_0$, $u(t)$ smooth and divergence-free for every $t \geq 0$ (and $M(t) \geq 0$).*

Theorem 7.4 (`ConditionalRegularity.conditional_regularity_of_open_hyps`)**.** *Under the hypotheses of Theorem 7.3 (stated for a time-dependent field $u_0(\cdot)$ that is divergence-free at all times, with $u_0(0)$ smooth and of finite energy), there is a u with $u(0) = u_0(0)$, $u(t)$ smooth and satisfying `satisfies_NavierStokes` (physical solution of the equations) for $t \geq 0$, and unique among smooth fields with these properties.*

Theorem 7.5 (`frohmanian_tether_theorem`)**.** *Assume: (hKato) a Kato local witness for every smooth divergence-free finite-energy datum and $\nu > 0$; (hDI) `FrohmanianTetherRate u₀ ν` for all u_0, ν; a uniform lower bound $\tau > 0$ on the witness times; and the integrability and two uniqueness hypotheses `hInt`, `huniq_shift`, `huniq` on existence intervals (Appendix A.3). Then there is a kernel $\mathfrak{T}_F$ such that: (1) for every F, ω satisfying the three hypotheses of Theorem 3.3, the tethered and classical brackets agree on (F, H); (2) every* antisymmetric *kernel satisfying (C1)–(C3), saturation and the polarization hypotheses equals $\mathfrak{T}_F$; (3) every kernel with density equal to the canonical tether density equals $\mathfrak{T}_F$; and (4) for every $\nu > 0$ and every smooth divergence-free finite-energy u_0 there are u, p with `NS_PDE u p ν`, $u(0) = u_0$, and $u(t)$ smooth and divergence-free for all $t \geq 0$.*

Remark 7.6. Clause (4) is a global-existence statement *conditional on* (hDI) for all data together with uniform Kato restart and uniqueness, all of which are open; it is not a regularity theorem. In clause (2), antisymmetry gives $B(F, F) = 0$, so saturation forces $T(\omega; F, F) = 0$ for all F, ω,

and then polarization of T forces $T \equiv 0$; since $T(\omega; F, F) = -\kappa\,\mathcal{E}(\omega)\int \|\Pi_u \delta F/\delta\omega\|^2$ is not identically zero in general, clause (2) is expected to apply to no kernel (this observation is ours and is not formalized).

Theorem 7.7 (Non-vacuity at zero data; `conditional_regularity_nonvacuous_at_zero`)**.** *For every $\nu > 0$ there is u with $u(0) = 0$, smooth and a physical solution for $t \geq 0$, unique among smooth physical solutions with zero data. Also `stepField_not_physical`: a time-discontinuous step field, which would satisfy `NS_PDE` because `deriv` is 0 at non-differentiable points, is excluded by `IsPhysicalNSSolution`.*

This shows only that the conclusion form of Theorem 7.4 is consistent in the trivial case; it gives no information on nonzero data.

8 The combined capstone

Theorem 8.1 (`Combined.combined_conditional_capstone`)**.** *Let a group `SDiff` act on `CoadjointOrbit`. Assume all of the following: `SDiffActionHyp SDiff`; `TetherIntegralPreserved SDiff`; `KineticEnergyDerivIsVelocity`; $\nabla \cdot (\mathrm{BS}\,\omega) = 0$ and $\|\mathrm{BS}\,\omega\|^2$ integrable for all ω; a trajectory $\omega(t)$ with `IsCoadjointMotion`; a symmetric kernel B with `h_proj`, `h_dens`, `h_weight`, `h_sat`, `h_nondeg` as in Theorem 5.3; u_0 smooth, divergence-free, of finite energy; $\nu > 0$; `KatoLocalWitness u₀ ν`; `FrohmanianTetherRate u₀ ν`; `FrohmanianTetherClosingBound ν`; `PhysicalBKMContinuationCriterionFor u₀ ν`; `KatoUniquenessOnExistenceTimes u₀ ν w`. Then: (1) T satisfies (C1), (C2), (C3); (2) $B = T$; (3) every Casimir is constant along $\omega(t)$; (4) $T(\omega; H, M) = 0$ for all ω, M; (5) `PhysicalVorticityCubicRate u₀ ν`, and there exist u, p with `NS_PDE u p ν`, $u(0) = u_0$, $u(t)$ smooth and divergence-free for all $t \geq 0$.*

Conclusion (4) is the obstruction: under the very hypotheses of the capstone the tether supplies no damping, so nothing in the geometric layer supports the hypotheses `FrohmanianTetherRate` and `FrohmanianTetherClosingBound`, which carry the entire analytic conclusion (5). The capstone also depends on Hypothesis 3.7, which we believe to be false (Section 3.1); if it is, the capstone is vacuous. No non-vacuity witness for the capstone is formalized.

9 The periodic Clay statement and a conditional reduction

`ClayStatements.Clay_B_exact` formalizes a version of Fefferman's statement (B) [4]: for every $\nu > 0$ and every smooth, divergence-free, $\mathbb{Z}^3$-periodic u_0 on $\mathbb{R}^3$ there is a solution (u, p) that is C^∞ on $[0, \infty) \times \mathbb{R}^3$, solves the equations with zero forcing, has $u(0) = u_0$, and is periodic for all $t \geq 0$. (Fefferman's formulation also contains a bounded-energy condition; the formal statement is as quoted in Appendix A.4.)

Theorem 9.1 (`clay_B_exact_of_periodic_tether_bound`)**.** *If `PeriodicTetherClosingBound ν` and `PeriodicBKMContinuation ν` hold for every $\nu > 0$, then `Clay_B_exact`.*

`PeriodicTetherClosingBound` is a uniform vorticity bound for all periodic smooth data and `PeriodicBKMContinuation` says that such a bound yields a global periodic smooth solution. Together they are at least as strong as the conclusion; the reduction is a formal bookkeeping step and makes no progress on (B).

10 What the Lean development does and does not prove

Proved (no further hypotheses). Symmetry of T and (C3); antisymmetry of $\{\cdot,\cdot\}_{\mathrm{cl}}$ and $\{H,H\}_{\mathrm{cl}} = 0$; Theorem 4.1 ($B(H,M) = 0$ for every symmetric B satisfying (C2)); `casimir_invariant` for the concrete permutation group; the comparison Lemma 7.1; the scaling identity and the scale non-invariance lemma; the local-maximum stretching bound `stretching_inner_le_at_max` (given its strain hypothesis `hCZ`); the zero-data facts and `stepField_not_physical`; the Fourier-symbol identities `PlanSteps.bsSymbol_div`, `PlanSteps.bsSymbol_curl` (mode-wise algebra, not connected to the real-space Biot–Savart integral).

Proved as implications from named, open hypotheses. (C2) for T (from Hypothesis 3.7 and integrability); (C1) for T (from `SDiffActionHyp` and `TetherIntegralPreserved`); canonicity (from saturation and polarization); Casimir conservation (from `IsCoadjointMotion`); the conditional Jacobiator lemma; the conditional regularity theorems 7.3, 7.4, 7.5 and the capstone 8.1 (from rate or closing-bound hypotheses, Kato existence, BKM continuation and uniqueness); the Clay (B) reduction.

Not proved, and not claimed.

- Global existence, smoothness or uniqueness for 3D Navier–Stokes with large data; any part of the Clay Millennium Problem.
- The cubic rate `FrohmanianTetherRate`, any closing bound, or any damping mechanism from the tether (its damping coefficient is 0 under (C2)).
- Hypothesis 3.7 ($\delta H/\delta\omega = u$), which we believe false as stated.
- That the Euler flow is a coadjoint motion; transitivity on Casimir levels; `TetherIntegralPreserved`.
- A Jacobi identity for $\{\cdot,\cdot\}_{\mathfrak{T}}$, or a Poisson structure; the Jacobi identity for $\{\cdot,\cdot\}_{\mathrm{cl}}$ (cited only).
- Kato local existence [6], Beale–Kato–Majda [3], parabolic smoothing, energy uniqueness: all are inputs, none is formalized.
- Anything on the quotient torus: `T3` is $\mathbb{R}^3$, and no diffeomorphism group of $\mathbb{T}^3$ is constructed.
- Derivation of T from (C1)–(C3): canonicity assumes diagonal agreement with T.

The physical motivations of earlier versions (fluid–gravity duality, Onsager's conjecture [5], metriplectic viscous completion) are not formalized and play no role in any statement here.

A Exact Lean statements

Quoted verbatim from the repository by `scripts/extract_statements.py`; proof bodies are replaced by `...`. Each block starts with its file and line.

A.1 Setting

```
-- FrohmanianTether/ArnoldGeometric.lean, line 172
public structure CoadjointOrbit where
  val : T3 → EuclideanSpace ℝ (Fin 3)
  property : ∀ x, NavierStokes3D.div val x = 0
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 313
public noncomputable def FunctionalDerivative (F : CoadjointOrbit → ℝ)
    (ω : CoadjointOrbit) : T3 → EuclideanSpace ℝ (Fin 3) :=
  if h : ∃ δ, IsGateauxRepresentative F ω δ then Classical.choose h else 0
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 915
public noncomputable def KineticEnergyHamiltonian (ω : CoadjointOrbit) : ℝ :=
  (1/2) * ∫ x, ‖BiotSavart ω.val x‖^2 ∂(volume)
```

```
-- FrohmanianTether/NS_Equations.lean, line 2149
@[expose] public def navier_stokes_eq (u : TimeDependentVelocity) (p : TimeDependentPressure)
    (ν : ℝ) : Prop :=
  ∀ t ≥ (0 : ℝ), ∀ x : T3,
    time_deriv u t x + convective (u t) (u t) x + pressureGradient (p t) x =
        ν • laplacian (u t) x ∧
      div (u t) x = 0
```

```
-- FrohmanianTether/NS_Equations.lean, line 2259
@[expose] public def IsPhysicalNSSolution (u : TimeDependentVelocity) : Prop :=
  (∀ t ≥ (0 : ℝ), ∀ x : T3, DifferentiableAt ℝ (fun s => u s x) t) ∧
  (∀ t ≥ (0 : ℝ), Continuous (u t)) ∧
  (∀ t ≥ (0 : ℝ), Integrable (fun x : T3 => ‖u t x‖ ^ 2)) ∧
  (∀ t ≥ (0 : ℝ), (∫ x, ‖u t x‖ ^ 2 ∂volume) ≤ ∫ x, ‖u 0 x‖ ^ 2 ∂volume) ∧
  (∀ t ≤ (0 : ℝ), u t = u 0)
```

```
-- FrohmanianTether/NS_Equations.lean, line 2422
public structure KatoLocalWitness (u₀ : VelocityField) (ν : ℝ) where
  T : ℝ
  Tpos : 0 < T
  u : TimeDependentVelocity
  p : TimeDependentPressure
  smooth : ∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (u t)
  init : u 0 = u₀
  pde : NS_PDE u p ν
  phys : IsPhysicalNSSolution u
```

A.2 Tether, (C1)–(C3), obstruction, canonicity

```
-- FrohmanianTether/ForMathlib/Projection.lean, line 35
@[expose] public noncomputable def Pi_u (u v : VelocityField) : VelocityField :=
  fun x => v x - (
    (∫ y, pairing (v y) (u y) ∂volume) /
    (∫ y, ‖u y‖ ^ 2 ∂volume)
  ) • u x
```

```
-- FrohmanianTether/SymplecticTether.lean, line 352
@[expose] public noncomputable def TetherKernel (ω : CoadjointOrbit) (F G : Functional) : ℝ :=
  -kappa * Enstrophy ω * ∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x)
    ∂volume
```

```
-- FrohmanianTether/SymplecticTether.lean, line 359
public noncomputable def TetheredBracket (F G : Functional) (ω : CoadjointOrbit) : ℝ :=
  -- The Frohmanian Symplectic Tether bracket: the classical Arnold Lie–Poisson bracket
  -- plus the minimal quadratic correction (the tether kernel) that satisfies (C1)–(C3).
  -- Canonicity is proved only inside the restricted kernel class and under the explicit
  -- hypotheses of `uniqueness_of_minimal_tether` (Uniqueness.lean); it is NOT an
  -- unconditional uniqueness statement. The correction is symmetric, so this bracket is
  -- not antisymmetric.
  ClassicalBracket F G ω + TetherKernel ω F G
```

```
-- FrohmanianTether/SymplecticTether.lean, line 376
@[expose] public def InvariantUnderCoadjointAction (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop
    ↪  :=
  -- (C1): The correction B is invariant under the coadjoint action of SDiff(T³).
  -- For all volume-preserving g, the value of the correction is unchanged when the vorticity
  -- and the test functionals are transformed by the coadjoint action.
  ∀ (g : SDiff) (F G : Functional) (ω : CoadjointOrbit),
    B (CoadjointAction g ω) F G = B ω F G
```

```
-- FrohmanianTether/SymplecticTether.lean, line 384
@[expose] public def DegenerateWRTKineticEnergy (B : CoadjointOrbit → Functional → Functional
↪  → ℝ) : Prop :=
  -- (C2): The correction is degenerate with respect to the kinetic-energy Hamiltonian H.
  -- The Hamiltonian vector field generated by H is exactly the classical one (no modification
  -- to the reversible Euler dynamics). This is the energy-preserving degeneracy on energy.
  ∀ (F : Functional) (ω : CoadjointOrbit),
    B ω F KineticEnergyHamiltonian = 0
```

```
-- FrohmanianTether/SymplecticTether.lean, line 392
@[expose] public def ProducesControllableNegativeFeedback
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (F : Functional) (ω : CoadjointOrbit),
    B ω F F ≤
      -kappa * Enstrophy ω * ∫ x,
          ‖Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x‖ ^ 2
        ∂volume
```

```
-- FrohmanianTether/SymplecticTether.lean, line 415
public theorem tetherKernel_comm (ω : CoadjointOrbit) (F G : Functional) :
    TetherKernel ω F G = TetherKernel ω G F := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 435
public theorem tetherKernel_C3 (F : Functional) (ω : CoadjointOrbit) :
    TetherKernel ω F F ≤
      -kappa * Enstrophy ω * ∫ x,
          ‖Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x‖ ^ 2
        ∂volume := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 1068
public theorem tetherKernel_degenerates_on_kinetic_energy
    (F : Functional) (ω : CoadjointOrbit)
    (hδH : FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω)
    (hdiv : ∀ x, div (velocity_from_vorticity ω) x = 0)
    (hInt : Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2)) :
    TetherKernel ω F KineticEnergyHamiltonian = 0 := ...
```

```
-- FrohmanianTether/PlanSteps.lean, line 65
def KineticEnergyDerivIsVelocity : Prop :=
  ∀ ω : CoadjointOrbit, FunctionalDerivative KineticEnergyHamiltonian ω =
  ↪  velocity_from_vorticity ω
```

```
-- FrohmanianTether/Combined.lean, line 51
theorem tetherKernel_C2_of_KEDeriv
    (hδ : PlanSteps.KineticEnergyDerivIsVelocity)
    (hdiv : ∀ ω x, div (velocity_from_vorticity ω) x = 0)
    (hInt : ∀ ω, Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2)) :
    DegenerateWRTKineticEnergy TetherKernel := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 220
public structure SDiffActionHyp (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
↪  :
    Prop where
  casimir_invariant : ∀ (f : ℝ → ℝ) (g : SDiff) (ω : CoadjointOrbit),
    (∫ x, f ‖(g • ω).val x‖ ∂volume) = ∫ x, f ‖ω.val x‖ ∂volume
  transitive_on_casimir_levels : ∀ ω₁ ω₂ : CoadjointOrbit,
    (∀ f : ℝ → ℝ, Casimir f ω₁ = Casimir f ω₂) → ∃ g : SDiff, g • ω₁ = ω₂
```

```
-- FrohmanianTether/SymplecticTether.lean, line 974
@[expose] public def TetherIntegralPreserved (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] : Prop :=
  ∀ (g : SDiff) (F G : Functional) (ω : CoadjointOrbit),
    (∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity (CoadjointAction g ω))
          (FunctionalDerivative F (CoadjointAction g ω)) x)
        (Pi_u (velocity_from_vorticity (CoadjointAction g ω))
          (FunctionalDerivative G (CoadjointAction g ω)) x) ∂volume) =
    ∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x) ∂volume
```

```
-- FrohmanianTether/SymplecticTether.lean, line 990
public theorem tether_coadjoint_invariance (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] (hyp : SDiffActionHyp SDiff)
    (h_density : TetherIntegralPreserved SDiff)
    (g : SDiff) (F G : Functional) (ω : CoadjointOrbit) :
    TetherKernel (CoadjointAction g ω) F G = TetherKernel ω F G := ...
```

```
-- FrohmanianTether/ConcreteSDiff.lean, line 85
theorem casimir_invariant (f : ℝ → ℝ) (g : G) (ω : CoadjointOrbit) :
    (∫ x, f ‖(g • ω).val x‖ ∂volume) = ∫ x, f ‖ω.val x‖ ∂volume := ...
```

```
-- FrohmanianTether/Combined.lean, line 63
theorem left_H_zero_of_C2_symm
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F) (hC2 : DegenerateWRTKineticEnergy B)
    (ω : CoadjointOrbit) (M : Functional) :
    B ω KineticEnergyHamiltonian M = 0 := ...
```

```
-- FrohmanianTether/Combined.lean, line 71
theorem tetherKernel_left_H_zero (hC2 : DegenerateWRTKineticEnergy TetherKernel)
    (ω : CoadjointOrbit) (M : Functional) :
    TetherKernel ω KineticEnergyHamiltonian M = 0 ∧ TetherKernel ω M KineticEnergyHamiltonian
    ↪  = 0 := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 1127
public theorem classicalBracket_antisymm (F G : CoadjointOrbit → ℝ) (ω : CoadjointOrbit) :
    ClassicalBracket F G ω = - ClassicalBracket G F ω := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 303
public def SaturatesTetherQuadratic
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (F : Functional) (ω : CoadjointOrbit), B ω F F = TetherKernel ω F F
```

```
-- FrohmanianTether/Uniqueness.lean, line 308
public def Polarizes
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (ω : CoadjointOrbit) (F G : Functional),
    2 * B ω F G =
      B ω (fun ω' => F ω' + G ω') (fun ω' => F ω' + G ω') -
        B ω F F - B ω G G
```

```
-- FrohmanianTether/Uniqueness.lean, line 449
@[expose] public def FactorsThroughProjection
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ ω : CoadjointOrbit, ∃ Q : (T3 → EuclideanSpace ℝ (Fin 3)) → (T3 → EuclideanSpace ℝ (Fin
  ↪  3)) → ℝ,
    (∀ a b, Q a b = Q b a) ∧
    ∀ F G : Functional, B ω F G =
      Q (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω))
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω))
```

```
-- FrohmanianTether/Uniqueness.lean, line 124
public def HasTetherKernelDensity
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (α : CoadjointOrbit → T3 → ℝ) : Prop :=
  ∀ (ω : CoadjointOrbit) (F G : Functional),
    B ω F G =
      ∫ x,
        α ω x *
          inner ℝ
            (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
            (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x)
        ∂volume
```

```
-- FrohmanianTether/Uniqueness.lean, line 605
public theorem uniqueness_of_minimal_tether
    (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (_hyp : SDiffActionHyp SDiff)
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (hC1 : InvariantUnderCoadjointAction SDiff B)
    (hC2 : DegenerateWRTKineticEnergy B)
    (hC3 : ProducesControllableNegativeFeedback B)
```

```
    (h_sat : SaturatesTetherQuadratic B)
    (h_polarB : Polarizes B)
    (h_polarT : Polarizes TetherKernel) :
    ∀ (ω : CoadjointOrbit) (F G : Functional), B ω F G = TetherKernel ω F G := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 583
public theorem uniqueness_of_minimal_tether_core
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (h_proj : ∀ ω (F F' G : Functional),
      Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) =
        Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F' ω) → B ω F G = B ω F' G)
    (h_density : FactorsThroughProjection B → ∃ α, HasTetherKernelDensity B α)
    (h_weight : ∀ α, HasTetherKernelDensity B α → ∃ c : ℝ, ∀ ω x, α ω x = -c * Enstrophy ω)
    (h_sat : SaturatesTetherQuadratic B)
    (h_nondeg : ∃ (ω : CoadjointOrbit) (F : Functional), Enstrophy ω * (∫ x,
        inner ℝ (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          ↪  ∂NavierStokes3D.volume) ≠ 0) :
    ∀ ω F G, B ω F G = TetherKernel ω F G := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 1172
public theorem tethered_jacobi_identity (F G H : Functional) (ω : CoadjointOrbit)
    (X Y Z : VelocityField)
    (_hF : FunctionalDerivative F ω = X)
    (_hG : FunctionalDerivative G ω = Y)
    (_hH : FunctionalDerivative H ω = Z)
    (hident : jacobiator F G H ω =
      ∫ x, cyclicLiePairing X Y Z x ∂volume)
    (hvanish : (∫ x, cyclicLiePairing X Y Z x ∂volume) = 0) :
    jacobiator F G H ω = 0 := ...
```

```
-- FrohmanianTether/PlanSteps.lean, line 69
def IsCoadjointMotion (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (ω : ℝ → CoadjointOrbit) : Prop :=
  ∃ g : ℝ → SDiff, ∀ t, ω t = g t • ω 0
```

```
-- FrohmanianTether/PlanSteps.lean, line 75
theorem casimir_conserved {SDiff : Type*} [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (H : SDiffActionHyp SDiff) (ω : ℝ → CoadjointOrbit) (hω : IsCoadjointMotion SDiff ω)
    (f : ℝ → ℝ) (t : ℝ) : Casimir f (ω t) = Casimir f (ω 0) := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 674
public theorem tethered_stretching_of_weight
    (ωField uField : VelocityField) (x : T3) (C M : ℝ)
    (hu : DifferentiableAt ℝ uField x)
    (hCZ : ‖fderiv ℝ uField x‖ ≤ C * vorticity_sup_norm ωField)
    (hmax : ‖ωField x‖ = M ∧ M = vorticity_sup_norm ωField)
    (orbit : CoadjointOrbit) (testF : Functional)
    (hweight : M ^ 4 ≤ Enstrophy orbit * ∫ y,
          ‖Pi_u (velocity_from_vorticity orbit)
              (FunctionalDerivative testF orbit) y‖ ^ 2 ∂volume) :
    inner ℝ (ωField x) (convective ωField uField x) +
        TetherKernel orbit testF testF ≤
      (C * M - kappa * M ^ 2) * M ^ 2 := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 644
public theorem reduced_pairing_of_unmodified_and_tether
```

```
    (u : TimeDependentVelocity) (p : TimeDependentPressure) (ν : ℝ)
    (t : ℝ) (_ht : 0 ≤ t) (x : T3)
    (_hNS : NS_PDE u p ν)
    (_hreg : VorticityTransportRegularity u p t x)
    (_hu : DifferentiableAt ℝ (u t) x)
    (C M : ℝ)
    (_hCZ : ‖fderiv ℝ (u t) x‖ ≤ C * vorticity_sup_norm (vorticity (u t)))
    (_htransp : inner ℝ (vorticity (u t) x)
      (convective (u t) (vorticity (u t)) x) = 0)
    (_hvisc : inner ℝ (vorticity (u t) x)
      (laplacian (vorticity (u t)) x) ≤ 0)
    (_hν : 0 ≤ ν)
    (_hmax : ‖vorticity (u t) x‖ = M ∧
      M = vorticity_sup_norm (vorticity (u t)))
    (_htether : ProducesControllableNegativeFeedback TetherKernel)
    (_hκ : kappa = CalderonZygmundConstant3D)
    (hcancels : inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) +
      kappa * M ^ 4 ≤ C * M ^ 3) :
    inner ℝ (vorticity (u t) x)
        (convective (vorticity (u t)) (u t) x) ≤
      (C * M - kappa * M ^ 2) * M ^ 2 := ...
```

```
-- FrohmanianTether/PlanSteps.lean, line 84
theorem tether_bound_not_scale_invariant (S C κ M : ℝ) (hκ : 0 < κ) (hM : M ≠ 0) :
    ¬ ∀ l : ℝ, 0 < l → l ^ 6 * S ≤ C * (l ^ 2 * M) ^ 3 - κ * (l ^ 2 * M) ^ 4 := ...
```

A.3 Analytic layer

```
-- FrohmanianTether/AnalyticPipeline.lean, line 85
public theorem comparison_ode_stability
    (C κ'' : ℝ) (hC : 0 < C) (hκ : 0 < κ'')
    (y : ℝ → ℝ)
    (hdiff : ∀ s ≥ (0 : ℝ), DifferentiableAt ℝ y s)
    (hy_dot : ∀ s ≥ (0 : ℝ), deriv y s ≤ C * (y s) ^ 2 - κ'' * (y s) ^ 3)
    (t : ℝ) (ht : 0 ≤ t) :
    y t ≤ max (y 0) (C / κ'') := ...
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 2223
public theorem riccati_ceiling_of_vorticity_di
    (u : TimeDependentVelocity) (C κ'' : ℝ)
    (hC : 0 < C) (hκ : 0 < κ'')
    (hdiff : ∀ s ≥ (0 : ℝ),
      DifferentiableAt ℝ (fun τ => vorticity_sup_norm (vorticity (u τ))) s)
    (hDI : ∀ s ≥ (0 : ℝ),
      deriv (fun τ => vorticity_sup_norm (vorticity (u τ))) s ≤
        C * vorticity_sup_norm (vorticity (u s)) ^ 2 -
          κ'' * vorticity_sup_norm (vorticity (u s)) ^ 3)
    (hcont : Continuous fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) :
    ∃ Y : ℝ,
      (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
      Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 737
@[expose] public abbrev FrohmanianTetherRate (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    0 < ν → NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u →
      ∃ C κ'' : ℝ, 0 < C ∧ 0 < κ'' ∧
```

```
        (∀ s ≥ (0 : ℝ),
          DifferentiableAt ℝ (fun τ => vorticity_sup_norm (vorticity (u τ))) s) ∧
        (∀ s ≥ (0 : ℝ),
          deriv (fun τ => vorticity_sup_norm (vorticity (u τ))) s ≤
            C * vorticity_sup_norm (vorticity (u s)) ^ 2 -
              κ'' * vorticity_sup_norm (vorticity (u s)) ^ 3) ∧
        Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) ∧
        (∀ τ ≥ (0 : ℝ), BddAbove (Set.range fun x => ‖vorticity (u τ) x‖))
```

```
-- FrohmanianTether/SymplecticTether.lean, line 725
@[expose] public abbrev FrohmanianTetherBound (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u →
      ∃ Y : ℝ,
        (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
        Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) ∧
        (∀ τ ≥ (0 : ℝ), BddAbove (Set.range fun x => ‖vorticity (u τ) x‖))
```

```
-- FrohmanianTether/ClayStatements.lean, line 47
@[expose] public def FrohmanianTetherClosingBound (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) →
        FrohmanianTetherBound u₀ ν
```

```
-- FrohmanianTether/NS_Equations.lean, line 3319
@[expose] public def BKMContinuationCriterion (u : TimeDependentVelocity) (ν : ℝ) : Prop :=
  ∀ T : ℝ, Tstar (u 0) ν = (T : EReal) →
    ¬ IntegrableOn (fun t => vorticity_sup_norm (vorticity (u t))) (Set.Icc 0 T)
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 1916
@[expose] public def PhysicalBKMContinuationCriterionFor (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u → BKMContinuationCriterion u ν
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 1930
@[expose] public def KatoUniquenessOnExistenceTimes (u₀ : VelocityField) (ν : ℝ)
    (w : KatoLocalWitness u₀ ν) : Prop :=
  ∀ (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
    T' ∈ existenceTimes u₀ ν → v 0 = u₀ → NS_PDE v q ν →
    (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
    IsPhysicalNSSolution v →
    ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t
```

```
-- FrohmanianTether/ClosureBlueprint.lean, line 262
public theorem conditional_regularity_of_open_hyps
    (u₀ : VelocityField) (ν : ℝ)
    (h_divfree : ∀ x, div u₀ x = 0)
    (h_smooth : ContDiff ℝ (⊤ : ℕ∞) u₀)
    (h_finite_energy : Integrable (fun x : T3 => ‖u₀ x‖ ^ 2))
    (h_pos_ν : ν > 0)
    (w : KatoLocalWitness u₀ ν)
    (hRiccati : FrohmanianTetherBound u₀ ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor u₀ ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes u₀ ν w) :
    ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
      NS_PDE u p ν ∧
```

```
      u 0 = u₀ ∧
      (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
      (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0) := ...
```

```
-- FrohmanianTether/ConditionalRegularity.lean, line 207
public theorem conditional_regularity_of_open_hyps
    (u0 : TimeDependentVelocity)
    (h_div_free : ∀ t x, div (u0 t) x = 0)
    (h_smooth : IsSmooth (u0 0))
    (h_finite_energy : Integrable (fun x : T3 => ‖u0 0 x‖ ^ 2))
    (ν : ℝ) (h_ν_pos : ν > 0)
    (w : KatoLocalWitness (u0 0) ν)
    (hRiccati : FrohmanianTetherBound (u0 0) ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor (u0 0) ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes (u0 0) ν w) :
  ∃ (u : TimeDependentVelocity),
    (∀ t ≥ (0 : ℝ), IsSmooth (u t)) ∧
    (u 0 = u0 0) ∧
    (∀ t ≥ (0 : ℝ), satisfies_NavierStokes u ν) ∧
    (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
    (∀ v : TimeDependentVelocity,
      (∀ t ≥ (0 : ℝ), IsSmooth (v t)) →
      v 0 = u0 0 →
      (∀ t ≥ (0 : ℝ), satisfies_NavierStokes v ν) →
      ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

```
-- FrohmanianTether/ConditionalRegularity.lean, line 79
public theorem frohmanian_tether_theorem
    (hKato : ∀ u₀ ν, ContDiff ℝ (⊤ : ℕ∞) u₀ → (∀ x, div u₀ x = 0) →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) → 0 < ν →
      KatoLocalWitness u₀ ν)
    (hDI : ∀ u₀ ν, FrohmanianTetherRate u₀ ν)
    (τ : ℝ) (hτ : 0 < τ)
    (hTlb : ∀ u₁ ν (hsm : ContDiff ℝ (⊤ : ℕ∞) u₁)
      (hdiv : ∀ x, div u₁ x = 0)
      (hE : Integrable (fun x : T3 => ‖u₁ x‖ ^ 2))
      (hν : 0 < ν),
      τ ≤ (hKato u₁ ν hsm hdiv hE hν).T)
    (hInt : ∀ u₀ ν (_hν : 0 < ν) (T : ℝ) (v : TimeDependentVelocity)
      (q : TimeDependentPressure),
      T ∈ existenceTimes u₀ ν →
      v 0 = u₀ →
      NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      IsPhysicalNSSolution v →
      ∀ t ∈ Set.Icc (0 : ℝ) T,
        Integrable (fun x : T3 => ‖v t x‖ ^ 2))
    (huniq_shift : ∀ u₀ ν (_hν : 0 < ν) (T : ℝ) (v : TimeDependentVelocity)
      (q : TimeDependentPressure),
      T ∈ existenceTimes u₀ ν →
      v 0 = u₀ →
      NS_PDE v q ν →
      (∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (v t)) →
      IsPhysicalNSSolution v →
      ∀ t ∈ Set.Icc (0 : ℝ) T,
        ∀ (w' : KatoLocalWitness (v t) ν),
          ∀ s ∈ Set.Icc (0 : ℝ) w'.T, v (t + s) = w'.u s)
    (huniq : ∀ u₀ ν (w : KatoLocalWitness u₀ ν),
      ∀ (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
        T' ∈ existenceTimes u₀ ν →
```

```
        v 0 = u₀ →
        NS_PDE v q ν →
        (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
        IsPhysicalNSSolution v →
        ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t) :
  ∃ (𝔗_F : CoadjointOrbit → Functional → Functional → ℝ),
    (∀ F ω,
      FunctionalDerivative KineticEnergyHamiltonian ω =
          velocity_from_vorticity ω →
      (∀ x, div (velocity_from_vorticity ω) x = 0) →
      Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2) →
      TetheredBracket F KineticEnergyHamiltonian ω =
        ClassicalBracket F KineticEnergyHamiltonian ω) ∧
    (∀ (SDiff : Type) [Group SDiff] [MulAction SDiff CoadjointOrbit] (B : CoadjointOrbit →
    ↪  Functional → Functional → ℝ),
      (∀ ω F G, B ω F G = -B ω G F) →
      InvariantUnderCoadjointAction SDiff B →
      DegenerateWRTKineticEnergy B →
      ProducesControllableNegativeFeedback B →
      SaturatesTetherQuadratic B →
      Polarizes B →
      Polarizes TetherKernel →
      ∀ ω F G, B ω F G = 𝔗_F ω F G) ∧
    (∀ (B : CoadjointOrbit → Functional → Functional → ℝ)
      (α : CoadjointOrbit → T3 → ℝ),
      HasTetherKernelDensity B α →
      (∀ ω x, α ω x = canonicalTetherDensity ω x) →
      ∀ ω F G, B ω F G = 𝔗_F ω F G) ∧
    (∀ (u₀ : VelocityField) (ν : ℝ),
      0 < ν →
      (∀ x, div u₀ x = 0) →
      ContDiff ℝ (⊤ : ℕ∞) u₀ →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) →
      ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
        NS_PDE u p ν ∧
        u 0 = u₀ ∧
        (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
        (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
        (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0)) := ...
```

```
-- FrohmanianTether/ClayStatements.lean, line 517
public theorem conditional_regularity_nonvacuous_at_zero (ν : ℝ) (hν : 0 < ν) :
    ∃ (u : TimeDependentVelocity),
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (u t)) ∧
      (u 0 = 0) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes u ν) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ v : TimeDependentVelocity,
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (v t)) →
        v 0 = 0 →
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes v ν) →
        ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

```
-- FrohmanianTether/ClayStatements.lean, line 396
public theorem stepField_not_physical : ¬ IsPhysicalNSSolution stepField := ...
```

```
-- FrohmanianTether/Combined.lean, line 87
theorem combined_conditional_capstone
    (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (hyp : SDiffActionHyp SDiff)
```

```
    (h_density : TetherIntegralPreserved SDiff)
    (hδ : PlanSteps.KineticEnergyDerivIsVelocity)
    (hdivω : ∀ ω x, div (velocity_from_vorticity ω) x = 0)
    (hIntω : ∀ ω, Integrable (fun y => ‖velocity_from_vorticity ω y‖ ^ 2))
    (ωt : ℝ → CoadjointOrbit) (hω : PlanSteps.IsCoadjointMotion SDiff ωt)
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (h_proj : ∀ ω (F F' G : Functional),
      Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) =
        Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F' ω) → B ω F G = B ω F' G)
    (h_dens : FactorsThroughProjection B → ∃ α, HasTetherKernelDensity B α)
    (h_weight : ∀ α, HasTetherKernelDensity B α → ∃ c : ℝ, ∀ ω x, α ω x = -c * Enstrophy ω)
    (h_sat : SaturatesTetherQuadratic B)
    (h_nondeg : ∃ (ω : CoadjointOrbit) (F : Functional), Enstrophy ω * (∫ x,
        inner ℝ (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          ↪  ∂NavierStokes3D.volume) ≠ 0)
    (u₀ : VelocityField) (ν : ℝ)
    (h_divfree : ∀ x, div u₀ x = 0)
    (h_smooth : ContDiff ℝ (⊤ : ℕ∞) u₀)
    (h_finite_energy : Integrable (fun x : T3 => ‖u₀ x‖ ^ 2))
    (h_pos_ν : ν > 0)
    (w : KatoLocalWitness u₀ ν)
    (hRate : FrohmanianTetherRate u₀ ν)
    (hClose : ClayStatements.FrohmanianTetherClosingBound ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor u₀ ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes u₀ ν w) :
    -- (1) C1, C2, C3 for the tether
    (InvariantUnderCoadjointAction SDiff TetherKernel ∧
      DegenerateWRTKineticEnergy TetherKernel ∧
      ProducesControllableNegativeFeedback TetherKernel) ∧
    -- (2) canonicity: B is the tether
    (∀ ω F G, B ω F G = TetherKernel ω F G) ∧
    -- (3) Casimirs conserved along the coadjoint motion
    (∀ f t, Casimir f (ωt t) = Casimir f (ωt 0)) ∧
    -- (4) no damping: T(M,H) = 0
    (∀ ω M, TetherKernel ω KineticEnergyHamiltonian M = 0) ∧
    -- (5) analytic layer, conditional on hRate / hClose
    TetheredLyapunov.PhysicalVorticityCubicRate u₀ ν ∧
    (∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
      NS_PDE u p ν ∧ u 0 = u₀ ∧
      (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
      (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0)) := ...
```

A.4 Periodic Clay statement

```
-- FrohmanianTether/ClayStatements.lean, line 250
@[expose] public def Clay_B_exact : Prop :=
  ∀ (ν : ℝ) (u₀ : VelocityField),
    0 < ν →
    ContDiff ℝ (⊤ : ℕ∞) u₀ →
    (∀ x, div u₀ x = 0) →
    IsZ3Periodic u₀ →
    ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
      IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

```
-- FrohmanianTether/ClayStatements.lean, line 568
@[expose] public def PeriodicTetherClosingBound (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
```

```
      ∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y
```

```
-- FrohmanianTether/ClayStatements.lean, line 580
@[expose] public def PeriodicBKMContinuation (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
      (∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y) →
        ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
          IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

```
-- FrohmanianTether/ClayStatements.lean, line 592
public theorem clay_B_exact_of_periodic_tether_bound
    (hclose : ∀ ν > (0 : ℝ), PeriodicTetherClosingBound ν)
    (hBKM : ∀ ν > (0 : ℝ), PeriodicBKMContinuation ν) : Clay_B_exact := ...
```

B Verification certificate

Repository	https://github.com/BenFrohman/frohmanian-tether-sdiff, release `v0.1.1`
Lean toolchain	`leanprover/lean4:v4.30.0-rc1`
Mathlib	`v4.30.0-rc1`, rev `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
Build	`lake build`: 2712 jobs, success
Modules	22 (`FrohmanianTether` and 21 submodules)
Theorems audited	519 (483 stated in the sources, 36 auto-generated)
Declarations audited	761 (theorems, definitions, structures, instances)
Axioms	only `propext`, `Classical.choice`, `Quot.sound`
`sorry` / `admit` / `native_decide`	none
Axiom declarations	none

The exact commit, the CI runs, and the full per-theorem axiom list are recorded in `CERTIFICATE.md` in the repository. Reproduction:

```
lake exe cache get
lake build
python3 scripts/check_no_sorry.py
lake env lean scripts/AxiomAudit.lean
```

`scripts/AxiomAudit.lean` computes `collectAxioms` for every non-internal declaration in the `FrohmanianTether` modules and raises an error (non-zero exit) if any axiom other than the three standard ones occurs (this includes `sorryAx`, produced by `sorry`, and `Lean.ofReduceBool`, produced by `native_decide`) or if any axiom is declared. `scripts/check_no_sorry.py` rejects the tokens `sorry`, `admit`, `native_decide` and `axiom` declarations outside comments and strings. Output for the headline declarations:

```
'Combined.combined_conditional_capstone' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
'Combined.left_H_zero_of_C2_symm' depends on axioms: [propext, Classical.choice, Quot.sound]
```

```
'Combined.tetherKernel_left_H_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'ConditionalRegularity.frohmanian_tether_theorem' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
AUDIT SUMMARY: theorems audited: 519 (483 stated in the sources, 36 auto-generated
  equation/congruence/injectivity lemmas); declarations audited: 761; violations: 0
AUDIT PASS: only propext, Classical.choice, Quot.sound; no sorryAx; no axiom declarations
```

Licence

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